Assess whether the existing NIR calibration for API concentration in a powder blend can be reused after changing the API and excipients, and whether adding Raman would keep it valid.

DECISION STATUS **Provisional feasibility assessment**

Raman leads (25/36); provisional until 'optical properties fluorescence' is checked.

Raman, Raman probe · 25/36

- Documented comparison candidate; not a mandatory second method.
  `S01 p14-15` `REF D01` `R_Q1_R01`
- Broad target-signal link; no catalog entry = UNKNOWN, never exclusion.
  `S01 p4-5` `REF §3,5` `R_Q1_01`

Client premise not supported: Adding a second method does not restore the validity of the old calibration after a formulation change; a new matrix calibration is needed. `R_Q1_06`
`REF D01 risks, §5` `H01`

Recommended methods

#1
25 /36
range [8–32]

Raman · Raman probe

Confidence: low · ⚠ critical question open

Q1	6 /9	Q1b	4 /9
Q2	8 /9	Q3	7 /9
Q4	2 /3		

#2

24 /36

range [7–32]

MIR · ATR probe

Confidence: low | ⚠ critical question open

Q1	6 /9	Q1b	4 /9
Q2	7 /9	Q3	7 /9
Q4	1 /3		

#3

22 /36

range [5–32]

Fluorescence · fluorimetric probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	7 /9	Q3	7 /9
Q4	0 /3		

▼ More methods (2)

#4

22 /36

range [7–31]

NIR · diffuse reflection probe

Confidence: low | ⚠ critical question open

Q1	3 /9	Q1b	4 /9
Q2	8 /9	Q3	7 /9
Q4	2 /3		

#5
22 /36
range [5–32]

UV-Vis · diffuse reflection probe

Confidence: low | ⚠ critical question open

Q1	**4** /9	Q1b	**4** /9
Q2	**7** /9	Q3	**7** /9
Q4	**0** /3		

Questions for the customer

1. high `Sample · optical properties · fluorescence`

   Does the matrix fluoresce (known fluorescent components, colour)?

   ▼ Why

   if fluorescence=present: leader MIR

   `R_Q1_R02`

2. high `Sample · optical properties · opacity`

   Is the sample clear, turbid or opaque at the measuring point, and does it change?

   ▼ Why

   if opacity=clear: leader MIR

   `R_Q2_04`

3. high `Sample · sample state constraints · contact`

   May a probe touch the product at the measuring point?

   ▼ Why

   if contact=allowed: leader MIR; if contact=prohibited: leader MIR

   `R_Q2_03, R_Q2_06`

To improve accuracy, add the customer's answers to the text above and click Analyze again.

Ruled out (1)

UV_VIS · UV_LENS
This interface collects light emitted by the sample itself; the sample is not a light source. `R_Q2_02`

Decision path · factors of the leading candidate per stage, with source

Q1

MATCH **+2** Broad target-signal link; no catalog entry = UNKNOWN, never exclusion.
`S01 p4-5` `REF §3,5` R_Q1_01

UNKNOWN **0** Separation of the target signal from the interferences of this matrix. MATCH only where an evidence row names the page (selectivity_source); no spectra in the materials = UNKNOWN, never a MISMATCH and never 'no interferences'. `S01 p5,10-11` R_Q1_04

UNKNOWN **0** Generic row: a known matrix background must not cover the target signal. Unknown overlap = UNKNOWN. Replaced by the specific rows R_Q1_R02 (Raman fluorescence) and R_Q1_M03 (MIR water). Soft only; a hard exclusion needs a proven overlap (R_Q1_02). `S01 p5,11` `REF §5` R_Q1_05

UNKNOWN **0** Independent validation material must be established, not assumed.
`REF D01/D02 experiment` `S01 p5,10` R_Q1_07

MATCH **+2** Documented comparison candidate; not a mandatory second method.
`S01 p14-15` `REF D01` R_Q1_R01

UNKNOWN **0** Unknown excitation/background = UNKNOWN; fluorescence without a spectrum does not forbid Raman. `S01 p11` `REF §5` R_Q1_R02

Q1b

UNKNOWN **0** LOD/LOQ only from a scoped CapabilityProfile; the materials give none.
`spec 04 §1` `REF §3,5` R_Q1B_02

UNKNOWN **0** Achieved error vs allowed error needs a comparable metric; none in the materials.
`REF D01/D02 experiment` `spec 04` R_Q1B_04

UNKNOWN **0** A laboratory is not a known reference uncertainty. `REF D01,D02` `H01` R_Q1B_05

UNKNOWN **0** End-to-end latency needs a profile; spectral exposure alone is not the latency.
`spec 04 §1` R_Q1B_06

UNKNOWN **0** Both range ends must be inside the calibration/detector scope.
`REF D01,D02` `S01 p16,18` R_Q1B_08

UNKNOWN **0** Qualitative sensitivity rating: context only, never a score. `XLSX sensitivity + notes` R_Q1B_10

Q2

MATCH **+1** Documented Method-interface edge (S01 Table 1). `XLSX` `S01 p4-6` R_Q2_01

MATCH **+1** A form missing from the list is UNKNOWN; MISMATCH only for a constructive conflict (solid vs liquid-only cell). `XLSX` `CL10` R_Q2_02

UNKNOWN **0** Excludes only this pair; other interfaces stay. Taking a sample out is R_Q2_10 (one cause, one rule). `S01 p5,10` R_Q2_03

MATCH **+2** A surface or local signal is not the bulk composition of a heterogeneous stream without a sampling plan. `S01 p8,13-14,17-18` `REF D08` R_Q2_07

UNKNOWN **0** Particle/bubble/surface variation must be covered by a sampling or averaging plan. Our recommendation to average is not a plan (no MATCH from it). `REF D01,D02` `S01 p15,17` R_Q2_08

Q3

UNKNOWN **0** No numeric limit in the materials. Hot extrusion does not exclude MIR.
`S02 p14` `REF D07` R_Q3C_01

UNKNOWN **0** Fouling + contact interface needs a feasible cleaning/maintenance plan; a promise is not a match. `REF D02,D07` R_Q3C_04

MATCH **+2** Preference is soft; a lab interface stays valid for the feasibility experiment. `REF §3`
R_Q3C_05

UNKNOWN **0** Do not invent certification. `design comparator` R_Q3C_06

UNKNOWN **0** Long silica cables are documented; no universal allowed length.
`S02 p6-10,18` `S01 p15` R_Q3C_07

MATCH **+1** Generic family is a soft concept check; a real cable with proven mismatch is R_Q3P_03.
`S02 p6` `REF §5` R_Q3C_08

Risks and assumptions

high The customer's premise is not supported: adding Raman to NIR does not restore the validity of the old NIR calibration after the API and excipients changed. The existing calibration cannot be assumed valid for the new formulation. What still meets the goal: recalibrate NIR on the new formulation with designed batches and test Raman or MIR as comparison, or as a combination that must beat the best single method.

high Ranking vs assessment: The ranking (Raman 25, MIR 24, fluorescence 22) and the method B scores agree. The assessment is conditional, because fluorescence is a critical unknown: if the matrix fluoresces, the leader changes to MIR. The 1-point gap between Raman and MIR is far inside the overlapping intervals, so there is no real winner. Fluorescence is tied with NIR and UV-Vis at 22, and NIR, the customer's own method, sits just below the top 3 and should stay in the comparison. The fluorescence entry carries the mode 'raman_backscatter', which is probably a mapping error. The MIR entry is an ATR probe, which is doubtful for powder contact, and the diffuse-reflection variant (24, interval 10–31, no critical unknown) may fit a powder better. All sides rest on draft scoring, so this is for expert review.

high Fluorescence of the new API or excipients is unknown and could swamp the Raman signal, so the Raman lead is conditional. Check pure materials and blend first.

medium A probe at the feed frame sees only a local signal. Without a sampling plan and the powder's heterogeneity known, a local reading may not represent the bulk API content.

medium The API concentration range is unknown. A low API fraction would reduce the sensitivity of Raman and MIR, and with NIR-style overlapping bands the quantification would depend on the new calibration.

medium MIR ATR on a moving powder: contact, crystal fouling or attack, and the maximum fiber length and temperature are unconfirmed. The MIR link length and routing must be confirmed by AFS.

medium Raman laser can heat or damage dark or sensitive powders, and laser safety and explosion protection (ATEX) for the tablet area must be checked. Hazardous-area status is unknown.

low No reference method, accuracy need, sampling plan or analyzer distance is known, so acceptance criteria and fiber length cannot be fixed yet. Assumptions such as solid-powder phase and no water are not confirmed.

First feasibility experiment

Decide whether API concentration in the new powder blend can be measured reliably with fiber probes at the feed frame. Compare Raman and MIR against the customer's reference method, with a recalibrated NIR as baseline, and test whether adding Raman gives anything over the best single method.

Samples

At least 2 calibration batches plus 2 independent validation batches. Suggested 30–40 samples per batch across the API range, final numbers to agree with the customer. — Use the new API and excipients with spiked or designed blends. Vary API content independently of the excipient ratio, for example with a constant-sum mixture design, so API is not correlated with one excipient. Also vary particle size, bulk density or compaction, and moisture. Add a few samples that mimic a standardised recipe, which in production varies little. Include old-formulation blends only as a check of the customer's premise.

Reference method

The customer's assay on the same sample, for example HPLC-UV of a sampled blend. If none exists, options are HPLC-UV assay of withdrawn blend samples, UV assay after dissolution of tablets, or gravimetric spiking with known API content.

Methods compared

Raman, MIR, NIR

Interferents

excipient ratio (placebo and filler variation), particle size and bulk density, moisture, fluorescent or coloured components, sample heterogeneity (local vs bulk)

Duration

About 4–6 weeks for the lab study after samples and the reference method are available. An in-line campaign under GMP change control takes longer, typically several additional weeks to months.

1. Step 1: Clarify the target and agree the API range, accuracy need and reference method with the customer, and collect representative samples of the new formulation.
2. Step 2: Record Raman spectra (testing at least two laser wavelengths if available) on pure API, each excipient and the blend. Check fluorescence and which API bands are separable from excipient bands. Check MIR bands in the same way.
3. Step 3: Measure the designed static samples with fiber probes in a lab setup that mimics the feed frame geometry (Raman probe, MIR ATR or diffuse-reflection probe, and NIR as baseline). Use repeated presentations, repacking and several positions to assess local-vs-bulk representativeness.
4. Step 4: Build calibrations on at least two batches per method (NIR recalibrated on the new formulation, Raman, MIR). Add a Raman+NIR model and check whether it beats the best single method.
5. Step 5: Validate on independent batches that were not used in calibration, split by batch and not by neighbouring spectra of one sample. Compare against the reference method.
6. Step 6: Only after a successful lab study, plan an in-line campaign at the feed frame under change control, with a sampling plan for reference samples.

Success criterion: Agree with the customer before the study. At minimum: on independent validation batches the RMSEP against the reference stays within the agreed accuracy, and the mean bias is not significantly different from zero (95% confidence interval within the agreed limit). The Raman+NIR combination counts only if it beats the best single method on the same samples.

Parameters and attributes · 38 Facts

Facts

Value · unit · status. Hover a row for the customer's words. Edit → Recalculate.

Fact	Value	Unit	Status
MeasurementTask · measurement goal	quantify		reported „…“
MeasurementTask · target	api		reported „…“
MeasurementTask · target definition	defined		reported „…“
MeasurementTask · spatial scope	local		inferred
MeasurementTask · sampling plan	—		unknown
MeasurementTask · reference method	—		unknown
MeasurementTask · performance requirements	—		unknown

Fact	Value	Unit	Status
MeasurementTask · time requirements	—		unknown
MeasurementTask · target range	—		unknown
MeasurementTask · business goal	pilot		reported „…“
Sample · matrix	pharma_blend		reported „…“
Sample · phase	solid		inferred
Sample · physical form	powder		reported „…“
Sample · optical properties · opacity	—		unknown
Sample · optical properties · fluorescence	—		unknown
Sample · heterogeneity	—		unknown
Sample · variability · formulation	changed		reported „…“
Sample · variability · physical	—		unknown
Sample · sample state constraints · contact	—		unknown
Sample · sample state constraints · sampling	—		unknown
Sample · relevant components · fluorophore	—		unknown
Sample · relevant components · absorber	—		unknown
ProcessContext · deployment preference	—		unknown
ProcessContext · temperature range	—		unknown
ProcessContext · temperature c	—	degC	unknown
ProcessContext · operation constraints · fouling	—		unknown
ProcessContext · operation constraints · hazardous	—		unknown

Fact	Value	Unit	Status
ProcessContext · instrument distance	—		unknown
ProcessContext · assets · fiber	—		unknown
ProcessContext · assets · fiber profile	—		unknown
ProcessContext · assets · must reuse	—		unknown
ProcessContext · assets · calibration scope	outside		inferred
ProcessContext · assets · method	NIR		reported „…“
MeasurementTask · reference method · independent	—		unknown
CustomerRequest · justification	—		unknown
CustomerRequest · proxy allowed	—		unknown
CustomerRequest · asks	assessment		reported „…“
CustomerRequest · combination to keep calibration	yes		reported „…“